



Melodoom brief

Status **In progress** ▾

Timing Mar 28, 2026 to Mar 28, 2026

Owners Samia Haynes joshuadupati@gmail.com

Overview

Melodoom is an AI-powered virtual companion app that makes sustainability personal, playful, and culturally resonant. Inspired by the nostalgia of Tamagotchi, each user adopts a digital music icon — a living, reactive character whose health, mood, and evolution are tied directly to the user's carbon, financial, and digital footprint. The better your choices, the more your artist thrives.

Objectives

Here are Melodoom's project objectives:

User Experience & Engagement

- Design an emotionally engaging companion experience where user behavior visibly and immediately affects their music icon's mood, health, and evolution

- Leverage music culture and nostalgia to make sustainability feel personal and fun rather than obligatory
- Deliver real-time feedback loops that reward positive habits and create meaningful stakes for negative ones

Sustainability Impact

- Enable users to track their carbon, financial, and digital footprint through simple, low-friction daily inputs
- Aggregate anonymized user data to generate neighborhood and campus-level sustainability trend reports
- Drive measurable behavioral change by connecting abstract environmental data to an outcome users are emotionally invested in

Technical Objectives

- Integrate Gemini API to power dynamic character behavior, personalized feedback, and footprint analysis
- Use Notion as a flexible, collaborative backend for user data, sustainability metrics, and community reporting (*planned for post-hackathon implementation*)
- Build a single-screen companion interface with responsive character states tied to real-time data inputs
- Architect the system to support future scaling across multiple user communities, institutions, and music icon libraries

Development Process

- The team began by locking the core data model and Gemini prompt before writing any UI code — the JSON output schema was tested in AI Studio until it returned clean structured responses every time
- SwiftUI was chosen for its native animation support and iOS 26 compatibility, allowing mood-state transitions (saturation, scale pulse, color multiply) to be implemented without external libraries
- The character display was built around a transparent PNG sprite with SwiftUI modifiers handling all three mood states — eliminating the need for multiple image assets
- Input screens were designed with tap options and sliders rather than text fields to minimize friction and maximize completion rate during the demo
- The current build consists of five screens — Home, Carbon, Finance, Digital, and Profile — each serving a distinct role in the user's daily check-in flow

- The team applied a strict cut list throughout: the Notion backend, character library, community dashboard, and user auth were all deferred to keep the 10-hour build focused on the core user experience
- Backend integration with Notion and full data persistence are scoped for the next development phase following the hackathon

Challenges

- Gemini prompt reliability was the highest-risk item — early testing returned inconsistent JSON structure, solved by enforcing a strict system instruction with explicit output format requirements
- Balancing the Pokémon aesthetic with a clean iOS feel required a deliberate split: UI for the home screen, minimal dark-mode cards for the input screens
- Keeping scope tight under time pressure required the team to make hard cuts early and revisit them only if time allowed

Outcome

- Melodoom delivers a fully functional single-screen iOS app where a music icon character reacts in real time to a user's daily carbon, financial, and digital footprint
- The Gemini API powers the character's dialogue and mood determination, Notion logs every session, and SwiftUI handles all visual state transitions natively
- The demo shows a clear before-and-after contrast: When Drake is thriving he smiles and celebrate, then if his mood continues to decrease he becomes sad before a bad day triggers the disapproving side-eye